

7 (a) (i) charge / potential (difference) or charge per (unit) potential (difference) B1 [1]

(ii) $(V = Q/4\pi\epsilon_0 r \text{ and } C = Q/V)$

for sphere, $C (= Q/V) = 4\pi\epsilon_0 r$ C1

$C = 4\pi \times 8.85 \times 10^{-12} \times 12.5 \times 10^{-2} = 1.4 \times 10^{-11} \text{ F}$ A1 [2]

(b) (i) $1/C_T = 1/3.0 + 1/6.0$

$C_T = 2.0 \mu\text{F}$ A1 [1]

(ii) total charge = charge on $3.0 \mu\text{F}$ capacitor = $2.0 (\mu) \times 9.0 = 18 (\mu\text{C})$ C1

potential difference = $Q/C = 18 (\mu)\text{C} / 3.0 (\mu)\text{F} = 6.0 \text{ V}$ A1 [2]

or

argument based on equal charges:

$3.0 \times V = 6.0 \times (9.0 - V)$ (C1)

$V = 6.0 \text{ V}$ (A1)

(iii) potential difference (= $9.0 - 6.0$) = 3.0 V C1

charge (= $3.0 \times 2.0 (\mu)$) = $6.0 \mu\text{C}$ A1 [2]

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